

MASTER TECHNICAL REFERENCE & VISUAL GUIDE

Computer Hardware Devices & Architecture

Comprehensive Technical Blueprint: Microprocessors, Memory Subsystems, Storage, I/O & System Interconnects

DOMAIN: Computer Engineering

SCOPE: Component Level to System Bus

STANDARDS: PCIe 5.0, DDR5, NVMe 2.0, USB4

TOPICS: CPU, GPU, RAM, SSD, Chipset, Motherboard



Figure 1: Core Computer Hardware & Peripheral Taxonomy (Processing, Memory, Storage, I/O & Expansion)



Figure 2: Disassembled Modern Desktop Hardware Subsystems (CPU, Motherboard, GPU, DDR5 RAM, NVMe SSD, PSU, Cooler)

01 Computer Architecture Foundations & System Buses

Von Neumann vs Harvard Architecture

Foundation

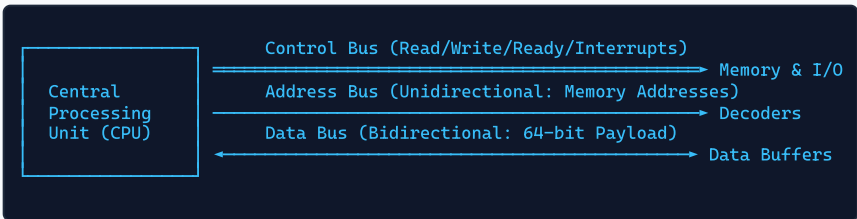
The structural paradigm of how processors interact with memory:

Feature	Von Neumann Architecture	Harvard Architecture
Memory Space	Unified memory (shared data & instructions)	Separate physical memories for code & data
Bus Structure	Single shared bus (Data + Instruction)	Independent instruction and data buses
Bottleneck	<i>Von Neumann Bottleneck</i> (cannot read code and data simultaneously)	No bus conflict; dual concurrent fetches
Modern Usage	Main PC RAM & System Architecture	L1 CPU Caches (L1-I and L1-D), DSPs, Microcontrollers

Modern Hybrid Model: Desktop/Server CPUs (Intel/AMD/ARM) operate as *Modified Harvard*: Unified external RAM, but separate internal L1 Instruction and L1 Data caches.

The Three System Buses

Interconnect



- Address Bus:** Unidirectional. Width defines max addressable physical RAM (2^N). A 32-bit bus addresses $2^{32} = 4 \text{ GB}$; a 48-bit physical address bus addresses $2^{48} = 256 \text{ TB}$.
- Data Bus:** Bidirectional. Carries actual data and instructions. Width defines word size transfer per cycle (64 bits = 8 bytes per transfer in modern CPUs).
- Control Bus:** Coordinates timing and direction signals (**MEMR**, **MEMW**, **IOR**, **IOW**, clock, bus requests, and hardware interrupts).

02 Central Processing Unit (CPU) Microarchitecture

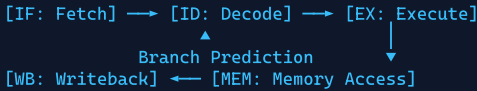
Core Execution Engine

Silicon

- ALU (Arithmetic Logic Unit):** Computes integer arithmetic (+ , - , *) and boolean logic (AND , OR , XOR).
- FPU / Vector Unit:** Floating Point Unit (IEEE 754) & SIMD engines (AVX2, AVX-512).
- Control Unit (CU):** Decodes opcodes, manages execution micro-ops, directs data paths.
- Registers:** Fastest on-chip storage (<1 cycle). General Purpose, Program Counter (PC/IP), Stack Pointer (SP), Flags/Status.

Instruction Pipeline & Hazards

Pipelining



- Structural Hazard:** Hardware resource conflict.
- Data Hazard (RAW/WAR/WAW):** Instruction depends on previous result (solved by Forwarding / Out-of-Order).
- Control Hazard:** Branch jumps (mitigated by Branch Predictors & Speculation).

ISA: x86-64 vs ARM64

Architecture

Parameter	x86-64 (CISC)	ARM64 (RISC)
Instruction Length	Variable (1–15 bytes)	Fixed (32 bits / 4 bytes)
Design Philosophy	Complex operations, legacy compatibility	Load/Store, simplified instructions
Power Profile	Higher power, high peak single-core	High power efficiency (perf/watt)
Dominant Space	Desktops, High-End Servers	Mobile, Apple M-series, AWS Graviton

CPU Memory Hierarchy & Cache Performance Latency

Hierarchy

Hierarchy Level	Typical Capacity	Latency (Clock Cycles)	Access Time (ns)	Bandwidth	Technology
CPU Registers	~1–2 KB total	0.5 – 1 cycle	~0.25 ns	Multi-TB/s	Custom CMOS D-Flip-Flops
L1 Cache (I/D per core)	32 KB – 64 KB per core	4 – 5 cycles	~1.0 ns	~2–3 TB/s	Static RAM (SRAM, 6T cells)
L2 Cache (per core)	512 KB – 2 MB per core	12 – 14 cycles	~3.5 ns	~1 TB/s	Static RAM (SRAM, 6T cells)
L3 Cache (Shared LLC)	16 MB – 96 MB (3D V-Cache)	35 – 50 cycles	~12–15 ns	~300–500 GB/s	High-density SRAM
Main Memory (RAM)	16 GB – 128 GB+	150 – 250 cycles	~60–80 ns	~50–100 GB/s	Dynamic RAM (DRAM, 1T-1C)
NVMe PCIe SSD	500 GB – 4 TB+	~50,000+ cycles	~10–30 μs	~3.5–14 GB/s	3D NAND Flash Memory

03 Motherboard, Chipsets, Sockets & Expansion Buses



Figure 3: CPU Socket (LGA 1700) with Retention Arm, Surrounding Multi-Phase VRM Heatsinks & Dual DDR5 RAM Slots

Motherboard Form Factors & CPU Sockets

Sockets

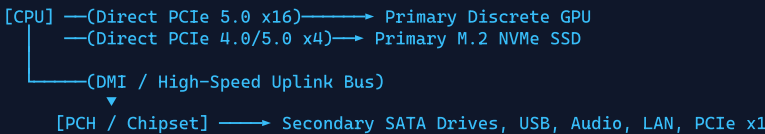
Form Factor	Dimensions	PCIe Slots	RAM Slots
Standard ATX	305 × 244 mm	3–4 slots	4 slots
Micro-ATX	244 × 244 mm	2–3 slots	2–4 slots
Mini-ITX	170 × 170 mm	1 slot	2 slots

- LGA (Land Grid Array):** Pins reside on the motherboard socket; CPU bottom has flat contact pads (Intel LGA 1700, AMD AM5).
- PGA (Pin Grid Array):** Pins are on CPU substrate; socket has insertion holes (AMD AM4).
- BGA (Ball Grid Array):** Soldered permanently via solder balls (Laptops, SoCs).

PCI Express (PCIe) Bandwidth & Chipset Topography

PCIe 5.0

PCIe Gen	Raw Transfer	x1 Bandwidth	x16 Bandwidth (GPU)
PCIe 3.0	8.0 GT/s	~0.985 GB/s	~15.75 GB/s
PCIe 4.0	16.0 GT/s	~1.969 GB/s	~31.51 GB/s
PCIe 5.0	32.0 GT/s	~3.938 GB/s	~63.02 GB/s
PCIe 6.0	64.0 GT/s	~7.56 GB/s	~121.0 GB/s



UEFI Firmware & Secure Boot: Modern 64-bit firmware replacing legacy BIOS. Supports GPT partition drives (>2.2 TB), graphical mouse UI, fast pre-boot bootloader signature verification via PKI certificates.

04 Primary Memory Architecture: RAM & ROM Technologies

DRAM vs SRAM & DDR Generations

RAM

- **SRAM (Static RAM):** 6 transistors per cell (flip-flop). No refresh cycle needed. Ultra-low latency, low density, high cost. Used in CPU caches.
- **DRAM (Dynamic RAM):** 1 transistor + 1 capacitor (1T-1C). Capacitor leaks charge; requires periodic electrical refresh (~every 64 ms). High density, used for main RAM.

Feature	DDR4	DDR5
Standard Data Rates	2133 – 3200 MT/s	4800 – 8400+ MT/s
Operating Voltage	1.2V	1.1V (on-module PMIC)
Channel Architecture	Single 64-bit channel / DIMM	Dual 32-bit subchannels / DIMM
Error Correction	Sideband ECC (Server only)	On-Die ECC (Standard) + Sideband
Burst Length	BL8	BL16 (Double prefetch)

Memory Timings, Multi-Channel & Virtual Memory

Performance

Latency & Timings (e.g., CL16-18-18-38):

- **CAS Latency (CL):** Number of clock cycles between memory controller issuing read request and first data bit output. Absolute latency: $t = \frac{\text{CL} \times 2000}{\text{Data Rate (MT/s)}} \text{ ns}$.
- **Multi-Channel Architecture:** Dual-channel doubles bus width from 64-bit to 128-bit, effectively doubling theoretical maximum peak bandwidth: $\text{Bandwidth (MB/s)} = \frac{\text{Clock (MHz)}}{8} \times \text{bytes} \times \text{Channels}$

MMU & Virtual Memory: Hardware Memory Management Unit (MMU) translates virtual addresses to physical RAM via Page Tables (4 KB standard pages). The **TLB (Translation Lookaside Buffer)** caches recent address mappings to prevent costly multi-level page table walks.

05 Secondary Storage: Magnetic HDDs vs Solid-State Drives



Figure 4: Physical Internal Comparison: Mechanical HDD (Spinning Magnetic Platter, Spindle, Voice-Coil Actuator Arm & Read/Write Head) vs. M.2 NVMe SSD (Flash Controller Microchip, DRAM Cache & 3D NAND Silicon Chips)

Hard Disk Drives (HDD)

Magnetic

- **Mechanical Components:** Rotating magnetic platters, spindle motor (5400 / 7200 RPM), voice-coil actuator arm with read/write heads.
- **Access Latency:** Seek time (3–10 ms) + Rotational latency (4.16 ms @ 7200 RPM).
- **Recording:** CMR (Conventional) vs SMR (Shingled - overlapping tracks, slower rewrite speeds).
- **Use Case:** Bulk cold storage & NAS archives.

NAND Flash Cell Types

Silicon

Cell	Bits/Cell	P/E Cycles	Endurance
SLC	1 bit	~100,000	Highest speed
MLC	2 bits	~3k–10k	High performance
TLC	3 bits	~1k–3k	Mainstream
QLC	4 bits	~100–1k	High capacity

NAND blocks must be erased before rewriting. Controlled via **Wear Leveling** and **TRIM** command.

SATA AHCI vs NVMe PCIe

Protocol

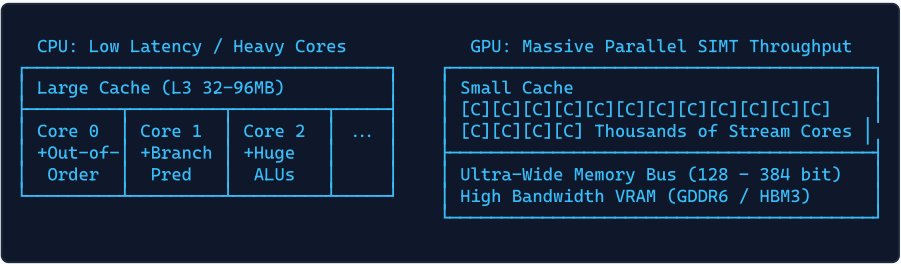
Feature	SATA III (AHCI)	NVMe (PCIe 4.0/5.0)
Max Bandwidth	600 MB/s (6 Gbps)	7,000–14,000 MB/s
Command Queues	1 queue	Up to 64,000 queues
Queue Depth	32 commands	64,000 per queue
Latency	~20–30 μs	~5–10 μs
CPU Overhead	High (Interrupts)	Low (Doorbell reg)

06 Graphics Processing Units (GPU) & Hardware Accelerators

CPU vs GPU Architecture Paradigms

Compute

Architectural design trade-offs: Latency Optimization vs Throughput Maximization:



- CPU:** Optimized for sequential code execution, branch-heavy logic, ultra-low single-thread latency.
- GPU:** SIMT (Single Instruction Multiple Threads). Executes identical instructions concurrently across massive matrices of vertices, pixels, or tensor weights.

Dedicated Processing Cores & VRAM

Acceleration

- Shader / CUDA / Stream Cores:** Standard FP32 & INT32 programmable arithmetic pipelines for 3D rasterization and physics.
- Ray Tracing (RT) Cores:** Dedicated silicon hardware designed to compute Bounding Volume Hierarchy (BVH) ray-triangle box intersections for photorealistic lighting and reflections.
- Tensor / Matrix Cores:** Specialized low-precision matrix multiply-accumulate engines (FP16, BF16, INT8, FP8) driving deep learning and AI inferences (DLSS, LLM inference).
- VRAM (Video Memory):**
 - GDDR6 / GDDR6X:** High-frequency discrete graphics memory operating across 128-bit to 384-bit buses (up to 1,000+ GB/s).
 - HBM (High Bandwidth Memory):** 3D-stacked silicon dies connected via silicon interposer directly adjacent to the GPU die (up to 3,000+ GB/s in datacenter GPUs like NVIDIA H100).

07 Power Supply Units (PSU) & Thermal Dissipation

Power Delivery & 80 PLUS Standards

Power

Converts high-voltage alternating current (110V/230V AC) to regulated DC power rails:

- +12V Rail:** Primary power rail for high-draw hardware (CPU & GPU).
- +5V & +3.3V Rails:** Powers motherboard logic, M.2 drives, USB devices.
- +5VSB (Standby):** Keeps motherboard listening for power button / WoL.

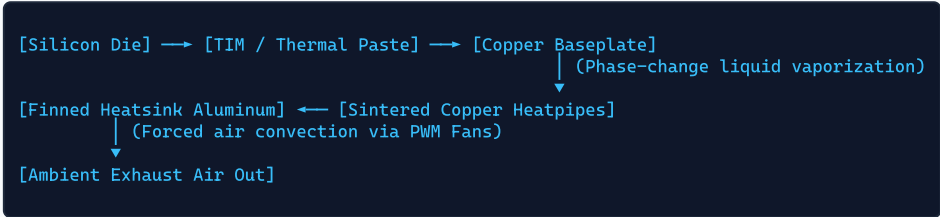
80 PLUS Rating	50% Load Efficiency	100% Load Efficiency
80 PLUS Bronze	85%	81%
80 PLUS Gold	90%	87%
80 PLUS Platinum	92%	89%
80 PLUS Titanium	94%	90%

ATX 3.0 & 12V-2x6 Connector: Supplies up to 600W through a single cable with sideband sense pins to prevent catastrophic GPU power-spike shutdowns.

Thermodynamics & Cooling Physics

Thermal

Cooling moves heat from die through three thermodynamic mechanisms:



- Air Cooler:** Heatpipes filled with low-pressure liquid vaporize heat to aluminum fin stacks. Reliable, zero pump failure risk.
- AIO Liquid Cooler:** Micro-channel copper pump block cycles liquid coolant to external radiator. Higher thermal buffer capacity.
- Thermal Throttling:** CPU/GPU automatically drops multiplier clocks once hitting maximum thermal junction ($T_{j\text{Max}} \approx 95^{\circ}\text{C} - 100^{\circ}\text{C}$) to prevent hardware damage.

08 Peripheral I/O Standards & Hardware Troubleshooting

Peripheral Interface Standards

Connectivity

Interface	Max Data Rate	Key Protocol / Features
USB 2.0	480 Mbps	Legacy peripherals (keyboards, mice)
USB 3.2 Gen 2	10 Gbps	SuperSpeed+ (External SSDs)
USB4 / Thunderbolt 4	40 Gbps	PCIe tunneling, DisplayPort Alt Mode, 100W+ PD
DisplayPort 2.1	80 Gbps (UHBR20)	Multi-Stream Transport (MST), DSC, 8K/16K
HDMI 2.1	48 Gbps (FRL)	Dynamic HDR, 4K@120Hz, eARC, VRR
Ethernet (RJ-45)	1G / 2.5G / 10 Gbps	802.3 Base-T Twisted pair (Cat 6/6a)
Wi-Fi 6E / Wi-Fi 7	9.6G / 46 Gbps	802.11ax / be: 6GHz band, 320MHz channels, MLO

Hardware Diagnostic & POST Flowchart

Diagnostics

When powering on a machine, the motherboard initiates **POST (Power-On Self-Test)**:

```
1. Press Power Button → PSU Power-Good (+12V stable)
2. [CPU LED / Code 00-15] → Detects CPU & microcode
   (If hangs: Check 8-pin CPU EPS power, bent socket pins)
3. [DRAM LED / Code 16-55] → Memory Training & SPD check
   (If hangs: Reseat RAM, test single DIMM in slot A2, clear CMOS)
4. [VGA LED / Code 60-99] → Initializes GPU & display output
   (If hangs: Reseat PCIe card, check PCIe auxiliary 8-pin cables)
5. [BOOT LED / Code A0-A9] → Bootloader handoff to NVMe/SATA
   (If hangs: UEFI boot order, partition table MBR vs GPT mismatch)
6. OS Kernel loads successfully!
```

Bottleneck Diagnosis: If GPU runs at 99-100% and CPU at 40%, system is *GPU-bound* (ideal for gaming). If CPU hits 100% while GPU hovers at 50-70%, system is *CPU-bound* (CPU cannot feed frame draw calls fast enough).